Annex II. Debt Sustainability Update

Based on preliminary data for 2015, Burkina Faso's risk of debt distress remains moderate, in line with the assessment at the time of the 2nd/3rd ECF reviews. Under the baseline scenario, all relevant debt ratios are projected to remain significantly below indicative thresholds over the projection period. Under the most extreme standardized stress test, the debt-to-exports ratio breaches the debt distress threshold around 2029. Although increasing at a faster pace, public debt is projected to remain below indicative thresholds. This debt profile is predicated on a prudent medium-term fiscal stance and relatively favorable prospects in the mining sector.

Background and underlying assumptions

A. Burkina Faso's public debt profile and evolution

1. **Based on preliminary data and estimates for 2015, Burkina Faso's stock of public debt has increased to 32.1 percent of GDP from 28.1 percent of GDP in 2014**. The stock of nominal debt increased by 10 percent despite initial efforts to maintain a tightened fiscal stance during the transition. External debt is estimated to drive the bulk of the increase of nominal debt as it increased by 12 percent while domestic debt only increased by 5 percent. Domestic debt remains low at 7 percent of GDP, and its share of total debt is gradually reverting back to historical levels (Text Table 1). However these trends are not expected to persist as Burkina Faso is making progress in tapping the domestic and regional debt market for its financing needs.

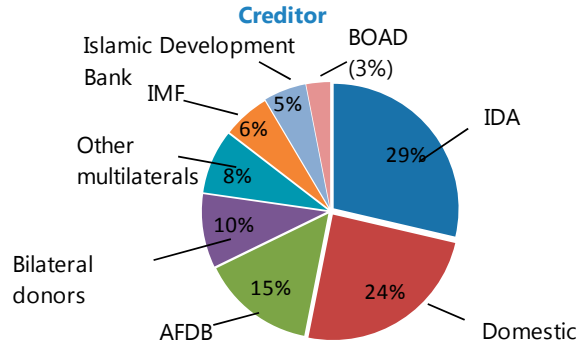
Text Table 1. Burkina Faso: Public Debt, 2011–15

	2011	2012	2013	2014	2015
	(US$ billions)				
Public debt	3.0	3.2	3.6	3.5	3.5
External	2.4	2.6	2.7	2.5	2.6
Domestic	0.6	0.7	0.9	1.0	1.0
	(CFAF billions)				
Public debt	1508	1616	1729	1863	2135
External	1216	1290	1291	1341	1561
Domestic	292	326	438	522	573
	(percent of GDP)				
Public debt	28.2	28.9	29.6	28.1	32.1
External	22.8	23.1	22.1	20.2	23.5
Domestic	5.5	5.8	7.5	7.9	8.6

Sources: Burkinabè authorities and IMF staff estimates

2. **The composition of debt is skewed towards more concessional debt.** IDA and the AFDB account for the largest shares with 29 and 15 percent of the total public debt stock, respectively, while other multilaterals (excluding the IMF) and bilateral donors account for 8 and 10 percent, respectively. The IMF holds 6 percent of Burkina Faso's total public debt stock and 8 percent of its external debt (Text Figure 1).

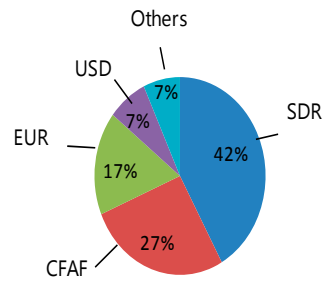
Text Figure 1. Burkina Faso: Public Debt by Creditor



Sources: Burkinabè authorities and IMF staff estimates.

3. **The stronger U.S. dollar in 2015 implied a moderate improvement in the external debt stock.** Although the nominal external debt stock in CFAF has increased by 18 percent, the stock of nominal external debt increased by only one percent as the appreciation of the U.S. dollar lowers the dollar value of all non-U.S. dollar denominated debt, which accounts for roughly three quarters of total debt (Text Figure 2).

Text Figure 2. Burkina Faso: Public Debt by Currency Denomination



Sources: Burkinabè authorities and IMF staff estimates.

B. DSA Assumptions

4. **The macroeconomic outlook is grimmer than envisaged at the time of the 2nd/3rd ECF reviews, owing to political uncertainty and weakening commodity prices** (Text Table 2).

- **Real GDP Growth** is projected at 5.4 percent on average over the medium term, down from 6 percent relative to the previous DSA amidst an uncertain political environment and the lingering impact of the recent terrorist attacks. Economic growth is projected to converge gradually to an average of 6 percent per year in the medium term, based on gradual improvements in energy supply, higher and better execution of public investment and the coming on stream of new gold mines.
- **Both gold and cotton prices are projected above their levels in the previous DSA, based on the latest WEO update** (March 4, 2016). Gold price projections are broadly in line with previous projections over the medium run but are projected to overshoot previous assumptions by 26 percent in the long term. Cotton price projections are initially lower but gradually increase by 43 percent above previous assumptions over the projection period.
- **The current account deficit is projected to average about 5.2 percent of GDP per year over the medium term** (compared to 7.7 percent of GDP in the previous DSA) reflecting the combined impact of higher gold and cotton prices, projected increase in gold production associated with two new opening mines and lower oil prices. The current account deficit is expected to stabilize around 7.5 percent of GDP over the long run.
- **The overall fiscal deficit** is projected to remain anchored at 3 percent of GDP over the medium term compared to an average 3.7 percent in the previous DSA but is projected to gradually converge towards about 6.1 percent of GDP as in the previous DSA.

Text Table 2. Burkina Faso: Changes in assumptions relative to the previous DSA

		2015	2016	2017	2018	2019	2020	2021
Gold (USD/ounce)	Current DSA (WEO)	1160	1219	1231	1234	1254	1274	1294
	2015 DSA	1180	1172	1187	1206	1226	1245	1265
Cotton prices (cts/lb)	Current DSA (WEO)	70	58	59	62	62	62	62
	2015 DSA	63	65	65	59	51	44	44
Real GDP growth (y/y)	Current DSA	4.0	5.2	5.9	6.1	6.2	6.0	6.0
	2015 DSA	5.0	6.0	6.5	6.6	6.6	6.6	6.6
Current account (% of GDP)	Current DSA	-6.4	-5.9	-5.3	-5.7	-6.0	-6.5	-7.2
	2015 DSA	-7.6	-7.8	-7.9	-8.1	-7.1	-7.2	-8.0
Overall fiscal balance (% of GDP)	Current DSA	-2.1	-3.1	-3.0	-3.2	-3.0	-3.1	-3.2
	2015 DSA	-2.5	-3.0	-3.7	-4.1	-4.1	-4.0	-4.0

Sources: World Economic Outlook, and IMF staff estimates and projections.

DSA Results

C. External Debt

5. **The baseline scenario remains broadly in line with that of the previous DSA, suggesting a moderate risk of debt distress.** The impact of the weaker growth outlook compared with the previous DSA on the debt metrics is being broadly balanced by that of higher mining exports and of the U.S. dollar appreciation. As in the previous DSA, the debt-to-exports ratio is projected to breach the relevant sustainability threshold under the most extreme standardized stress test scenario corresponding to a one-standard-deviation drop in exports growth relative to historical levels (the breach occurs in 2029 in the current DSA update compared with 2027 in the previous DSA).

6. **Although debt and debt service indicators remain significantly below relevant thresholds under the baseline scenario, there are risks attached to the relatively rapid pace of debt accumulation in the medium-term.** Mining revenues are inherently volatile and subject to uncertainties regarding both production volumes and prices, and the currently favorable U.S. dollar/euro exchange rate dynamics can reverse rapidly.

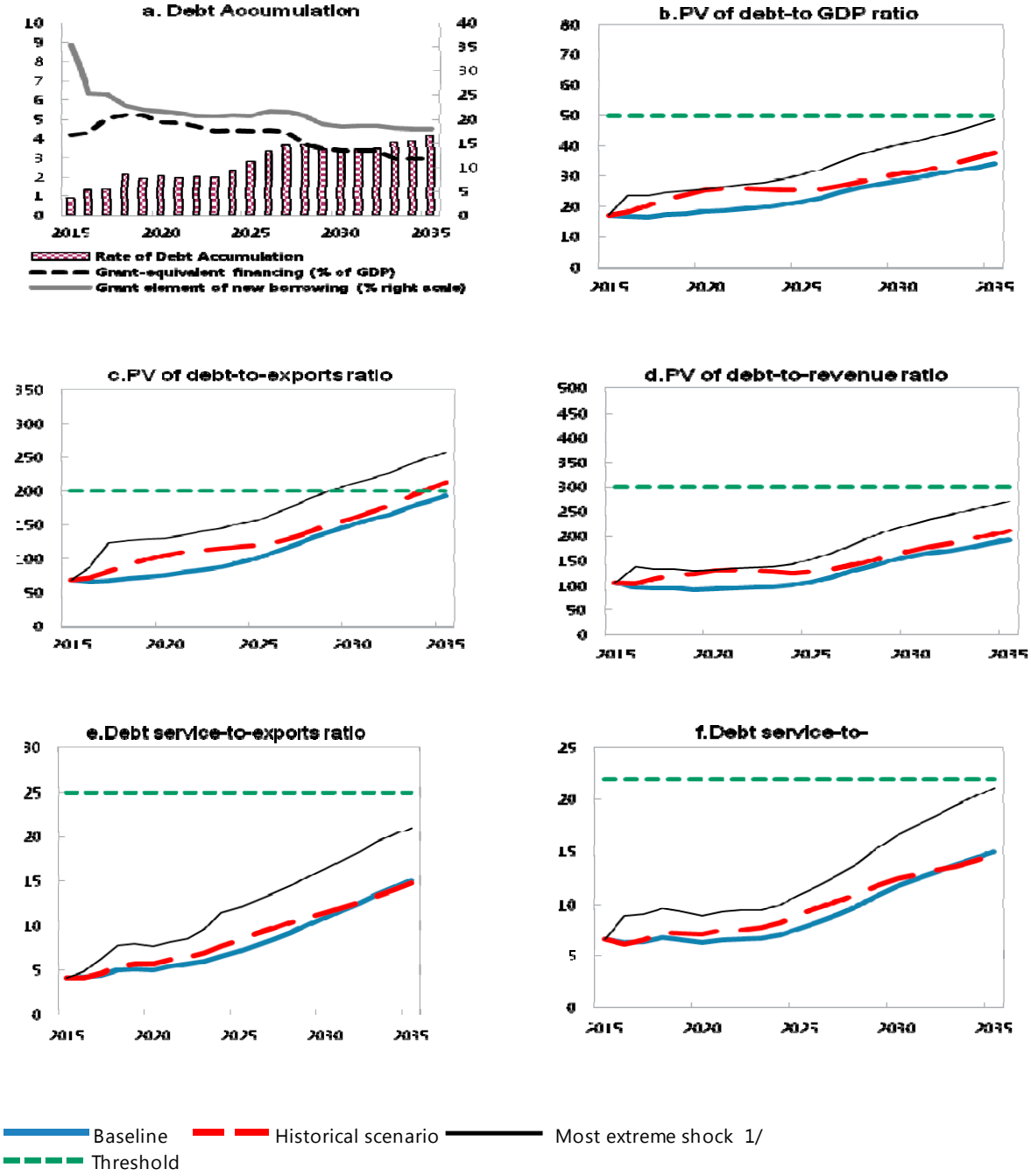
D. Total Public Debt

7. **Burkina Faso's total public debt is projected to increase more rapidly than in the previous DSA, although it would still remain below relevant thresholds throughout the projection period.** Total public debt is projected to rise from 32.2 percent of GDP in 2015 to 67.4 percent of GDP in 2035 (53 percent of GDP in the previous DSA), somewhat below the 74 percent threshold, while the stock of domestic debt gradually rises to 24.6 percent of GDP by the end of the projection period. The sharp build-up in public debt after the first decade of the projection period is based on the assumption that mining activity will slow down substantially, leading to a reduction of domestic revenue and more recourse to external and domestic financing. In the meantime the share of the non-concessional borrowing to total borrowing will increase from barely 29 percent to 55 percent at the end of the projection period. The authorities' heavy reliance on bank financing represents a downside risk, should the BCEAO need to tighten regional monetary conditions over the medium term.

Conclusion

The DSA results indicate that Burkina Faso's risk of debt distress remains "moderate." All relevant ratios remain below indicative thresholds under the baseline scenario. The DSA continues to underline the vulnerability of the debt path to shocks to exports as the ratio of debt to exports breaches its indicative threshold. In light of projected increases in domestic and external debt and vulnerabilities associated with export revenue and regional liquidity conditions, the authorities ought to strive to enhance debt management capacity, which is currently assessed as weak.

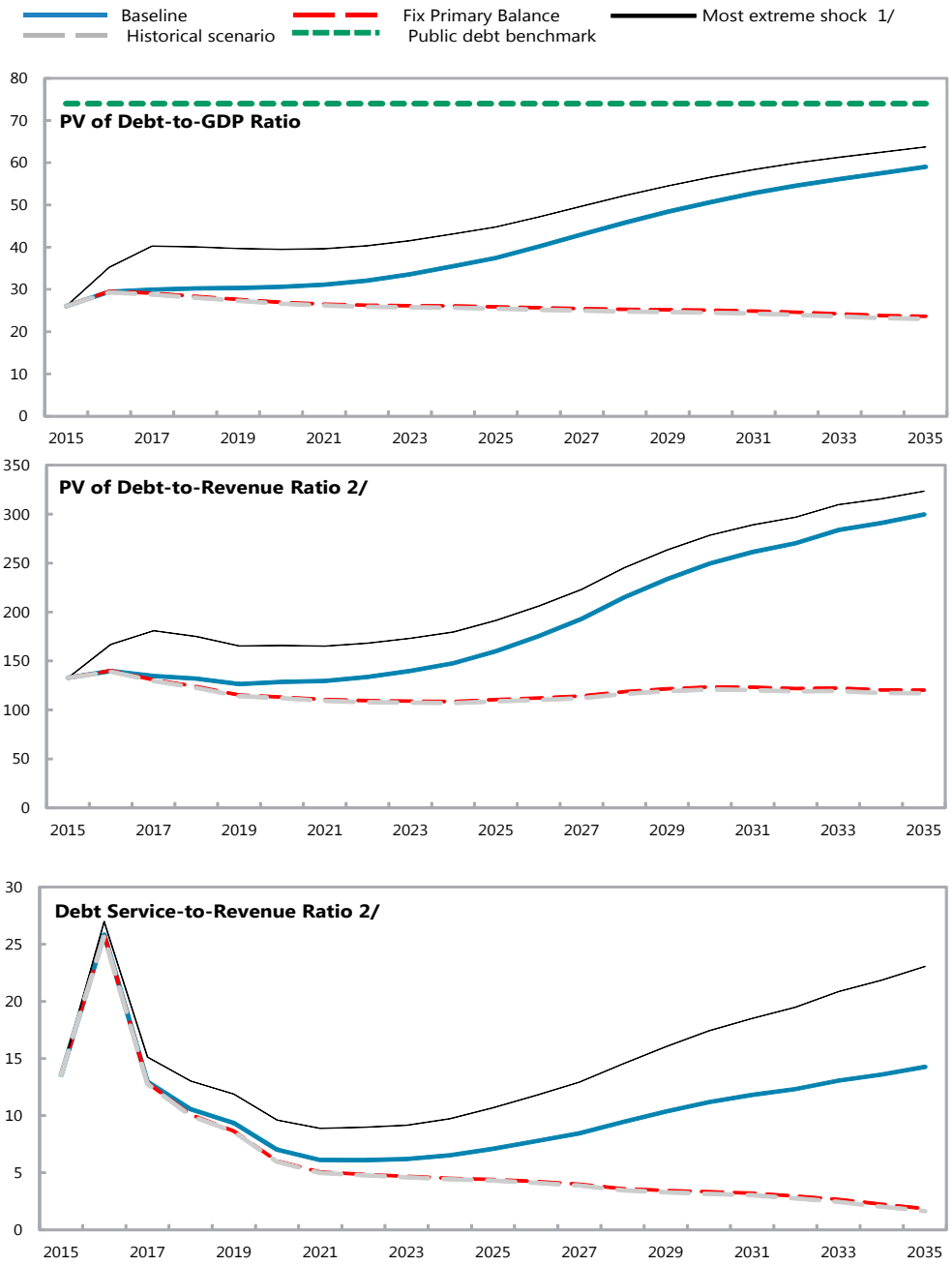
Figure A2.1. Burkina Faso: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2015–35 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure A2.2. Burkina Faso: Indicators of Public Debt Under Alternative Scenarios, 2015–35 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Table A2.1. Burkina Faso: External Debt Sustainability Framework, Baseline Scenario, 2014–35 1/

	Actual	Historical Average 6/	Standard Deviation 6/	Projections														
	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2021	2026	2030	2031	2032	2033	2034	2035
External debt (nominal) 1/	**20.2**			**23.5**	**23.6**	**23.1**	**23.7**	**23.9**	**24.3**		**24.6**	**29.3**	**36.0**	**37.2**	**38.4**	**39.8**	**41.2**	**42.8**
of which: public and publicly guaranteed (PPG)	20.2			23.5	23.6	23.1	23.7	23.9	24.3		24.6	29.3	36.0	37.2	38.4	39.8	41.2	42.8
Change in external debt	-1.9			3.3	0.1	-0.5	0.6	0.2	0.3		0.4	1.9	1.2	1.2	1.2	1.4	1.4	1.6
Identified net debt-creating flows	5.2			3.6	2.7	1.9	2.3	2.6	3.2		3.7	5.3	3.6	3.5	3.3	3.2	3.1	3.0
Non-interest current account deficit	**7.8**	**7.3**	**3.7**	**6.1**	**5.5**	**4.9**	**5.3**	**5.6**	**6.1**		**6.7**	**8.3**	**6.9**	**6.8**	**6.7**	**6.6**	**6.5**	**6.5**
Deficit in balance of goods and services	9.0			8.4	7.0	7.5	8.1	8.6	9.1		9.8	11.3	9.8	9.7	9.6	9.4	9.8	9.7
Exports	25.7			25.3	25.6	25.1	24.8	24.5	24.4		23.9	20.7	19.1	18.7	18.5	18.0	17.9	17.8
Imports	34.7			33.7	32.6	32.6	32.9	33.2	33.5		33.6	32.1	28.8	28.4	28.1	27.5	27.7	27.5
Net current transfers (negative = inflow)	-3.8	-4.8	0.8	-4.1	-3.1	-4.1	-4.1	-4.2	-3.9		-3.9	-3.4	-2.9	-2.9	-2.8	-2.7	-2.5	-2.4
of which: official	-1.8			-2.2	-1.3	-2.3	-2.3	-2.4	-2.2		-2.2	-1.6	-1.3	-1.2	-1.2	-1.1	-0.9	-0.9
Other current account flows (negative = net inflow)	2.7			1.8	1.6	1.5	1.3	1.1	1.0		0.8	0.3	0.0	0.0	-0.1	-0.1	-0.8	-0.8
Net FDI (negative = inflow)	**-2.3**	**-1.8**	**1.6**	**-1.9**	**-2.1**	**-2.1**	**-2.1**	**-2.1**	**-2.2**		**-2.2**	**-2.3**	**-2.4**	**-2.4**	**-2.4**	**-2.5**	**-2.5**	**-2.5**
Endogenous debt dynamics 2/	**-0.3**			**-0.6**	**-0.8**	**-0.9**	**-0.9**	**-0.9**	**-0.8**		**-0.8**	**-0.7**	**-0.9**	**-0.9**	**-0.9**	**-0.9**	**-1.0**	**-1.0**
Contribution from nominal interest rate	0.2			0.3	0.3	0.4	0.4	0.5	0.5		0.6	0.8	1.1	1.1	1.1	1.2	1.2	1.3
Contribution from real GDP growth	-0.9			-0.9	-1.1	-1.3	-1.3	-1.4	-1.3		-1.3	-1.5	-1.9	-2.0	-2.1	-2.1	-2.2	-2.3
Contribution from price and exchange rate changes	0.4			...	...	...	...	...	...		...	...	...	...	...	...	...	...
Residual (3-4) 3/ with changes in project grants	**-4.6**			**1.3**	**0.3**	**0.3**	**1.0**	**0.3**	**-0.3**		**-0.9**	**-1.2**	**-0.9**	**-0.7**	**-0.6**	**-0.7**	**-0.6**	**-0.4**
of which: exceptional financing	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PV of external debt 4/	15.1			17.3	16.8	16.8	17.5	17.9	18.4		18.9	23.1	28.6	29.6	30.7	31.9	33.1	34.4
In percent of exports	58.7			68.1	65.6	66.9	70.6	72.8	75.4		79.2	111.3	149.9	158.4	166.0	176.8	184.8	193.4
PV of PPG external debt	**15.1**			**17.3**	**16.8**	**16.8**	**17.5**	**17.9**	**18.4**		**18.9**	**23.1**	**28.6**	**29.6**	**30.7**	**31.9**	**33.1**	**34.4**
In percent of exports	**58.7**			**68.1**	**65.6**	**66.9**	**70.6**	**72.8**	**75.4**		**79.2**	**111.3**	**149.9**	**158.4**	**166.0**	**176.8**	**184.8**	**193.4**
In percent of government revenues	**87.5**			**107.3**	**97.2**	**94.7**	**95.0**	**92.0**	**94.0**		**95.4**	**118.4**	**160.4**	**167.1**	**172.4**	**178.7**	**185.5**	**192.8**
Debt service-to-exports ratio (in percent)	**3.1**			**4.2**	**4.3**	**4.5**	**5.1**	**5.2**	**5.0**		**5.5**	**7.8**	**10.9**	**11.8**	**12.6**	**13.6**	**14.3**	**15.1**
PPG debt service-to-exports ratio (in percent)	**3.1**			**4.2**	**4.3**	**4.5**	**5.1**	**5.2**	**5.0**		**5.5**	**7.8**	**10.9**	**11.8**	**12.6**	**13.6**	**14.3**	**15.1**
PPG debt service-to-revenue ratio (in percent)	**4.6**			**6.6**	**6.3**	**6.4**	**6.8**	**6.6**	**6.3**		**6.6**	**8.3**	**11.7**	**12.4**	**13.1**	**13.7**	**14.4**	**15.0**
Total gross financing need (Billions of U.S. dollars)	0.8			0.6	0.5	0.5	0.6	0.7	0.9		1.0	2.0	2.4	2.6	2.8	3.0	3.3	3.6
Non-interest current account deficit that stabilizes debt ratio	9.8			2.8	5.5	5.4	4.7	5.4	5.8		6.3	6.4	5.7	5.5	5.5	5.2	5.1	4.9
Key macroeconomic assumptions																		
Real GDP growth (in percent)	4.0	6.0	1.9	4.0	5.2	5.9	6.1	6.2	6.0	5.6	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
GDP deflator in US dollar terms (change in percent)	-1.6	3.9	6.9	-15.2	2.7	2.8	2.1	2.5	2.7	-0.4	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.2
Effective interest rate (percent) 5/	1.0	1.0	0.2	1.2	1.6	1.7	2.0	2.2	2.4	1.8	2.5	3.2	3.3	3.3	3.3	3.4	3.4	3.4
Growth of exports of G&S (US dollar terms, in percent)	1.7	21.6	26.4	-13.2	9.3	6.7	6.9	7.7	8.4	4.3	5.6	4.7	6.2	6.1	6.7	5.5	7.4	7.6
Growth of imports of G&S (US dollar terms, in percent)	-8.9	14.4	16.8	-14.2	4.4	9.0	9.1	9.9	9.9	4.6	8.7	6.8	6.6	6.4	6.9	5.9	9.2	7.3
Grant element of new public sector borrowing (in percent)	...	...	...	35.3	25.2	25.1	22.9	22.0	21.6	25.4	21.3	21.6	18.4	18.7	18.6	18.2	18.0	18.1
Government revenues (excluding grants, in percent of GDP)	17.3			16.1	17.3	17.8	18.4	19.4	19.6		19.8	19.5	17.8	17.7	17.8	17.8	17.8	17.8
Aid flows (in Billions of US dollars) 7/	0.6			0.5	0.5	0.6	0.7	0.8	0.8		0.8	1.0	1.0	1.1	1.1	1.0	1.1	1.2
of which: Grants	0.5			0.4	0.5	0.6	0.6	0.7	0.7		0.8	0.9	0.9	1.0	1.0	0.9	1.0	1.0
of which: Concessional loans	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2
Grant-equivalent financing (in percent of GDP) 8/	...			4.2	4.3	5.1	5.2	5.2	4.8		4.8	4.4	3.4	3.4	3.4	3.0	3.0	3.0
Grant-equivalent financing (in percent of external financing) 8/	...			77.0	73.6	75.4	67.7	69.7	67.4		67.7	54.2	46.0	45.4	44.4	39.5	38.8	37.1
Memorandum items:																		
Nominal GDP (Billions of US dollars)	12.5			11.0	11.9	12.9	14.0	15.3	16.6		18.0	26.5	36.3	39.2	42.4	45.8	49.6	53.7
Nominal dollar GDP growth	2.3			-11.8	8.0	8.8	8.4	8.8	8.8	5.2	8.1	8.1	8.1	8.1	8.1	8.1	8.1	8.3
PV of PPG external debt (in Billions of US dollars)	1.8			1.9	2.0	2.2	2.5	2.7	3.1		3.4	6.1	10.4	11.6	13.0	14.6	16.4	18.5
(PVt-PVt-1)/GDPt-1 (in percent)				0.9	1.3	1.4	2.2	2.0	2.1	1.6	2.0	3.4	3.4	3.5	3.5	3.8	3.9	4.2
Gross workers' remittances (Billions of US dollars)	0.2			0.2	0.2	0.2	0.3	0.3	0.3		0.3	0.5	0.6	0.6	0.7	0.7	0.8	0.8
PV of PPG external debt (in percent of GDP + remittances)	14.8			16.9	16.5	16.5	17.2	17.5	18.1		18.6	22.7	28.1	29.2	30.2	31.4	32.6	33.9
PV of PPG external debt (in percent of exports + remittances)	54.5			63.2	61.2	62.3	65.7	67.7	70.2		73.6	102.8	137.9	145.7	152.6	162.5	170.2	178.5
Debt service of PPG external debt (in percent of exports + remittances)	2.9			3.9	4.0	4.2	4.7	4.8	4.7		5.1	7.2	10.1	10.8	11.6	12.5	13.2	13.9

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2.2. Burkina Faso: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013–35

(In percent of GDP, unless otherwise indicated)

	Actual		Estimate	Projections									
	2013	2014	2015	2016	2017	2018	2019	2020	2021	2015-20 Average	2025	2030	2035
Public sector debt 1/	29.6	28.1	32.1	36.3	36.2	36.5	36.4	36.5	36.9		43.3	58.0	67.4
of which: foreign-currency denominated	22.1	20.2	23.5	23.6	23.1	23.7	23.9	24.3	24.6		27.4	36.0	42.8
Change in public sector debt	0.7	-1.5	4.1	4.1	0.0	0.2	0.0	0.0	0.4		2.2	2.4	1.7
Identified debt-creating flows	1.9	2.0	2.7	0.3	0.3	0.5	0.3	0.3	0.6		2.2	2.4	1.3
Primary deficit	3.9	-0.4	1.0	2.2	2.3	2.7	2.6	2.5	2.7	2.2	4.5	5.4	5.0
Revenue and grants	23.9	21.4	19.6	21.1	22.3	22.9	24.0	23.8	24.0		23.4	20.3	19.7
of which: grants	5.6	3.9	3.5	3.8	4.5	4.5	4.6	4.2	4.2		3.5	2.5	1.8
Primary (noninterest) expenditure	27.8	21.0	20.6	23.3	24.6	25.6	26.6	26.3	26.7		27.9	25.7	24.7
Automatic debt dynamics	-2.0	2.4	1.7	-2.0	-2.1	-2.2	-2.3	-2.2	-2.1		-2.2	-2.9	-3.6
Contribution from interest rate/growth differential	-1.1	0.0	-0.9	-1.4	-2.0	-2.1	-2.2	-2.1	-2.1		-2.2	-2.9	-3.6
of which: contribution from average real interest rate	0.7	1.1	0.2	0.2	0.0	-0.1	-0.1	-0.1	0.0		0.1	0.2	0.1
of which: contribution from real GDP growth	-1.8	-1.1	-1.1	-1.6	-2.0	-2.1	-2.1	-2.1	-2.1		-2.3	-3.1	-3.7
Contribution from real exchange rate depreciation	-0.9	2.4	2.6	-0.6	-0.1	-0.1	-0.2	0.0	0.0		...	...	...
Other identified debt-creating flows	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0
Residual, including asset changes	-1.2	-3.5	1.4	3.9	-0.3	-0.3	-0.3	-0.3	-0.2		0.0	0.0	0.3
Other Sustainability Indicators													
PV of public sector debt	...	23.6	26.1	29.4	30.0	30.2	30.3	30.6	31.1		37.5	50.7	59.0
of which: foreign-currency denominated	...	15.1	17.3	16.8	16.8	17.5	17.9	18.4	18.9		21.6	28.6	34.4
of which: external	...	15.1	17.3	16.8	16.8	17.5	17.9	18.4	18.9		21.6	28.6	34.4
PV of contingent liabilities (not included in public sector debt)	...	...	...	...	...	...	...	...	...		...	...	...
Gross financing need 2/	4.8	3.6	4.1	7.6	5.2	5.1	4.7	4.2	4.2		6.1	7.7	7.8
PV of public sector debt-to-revenue and grants ratio (in percent)	...	110.1	132.8	139.6	134.6	131.9	126.3	128.6	129.7		160.2	249.7	299.7
PV of public sector debt-to-revenue ratio (in percent)	...	134.2	161.3	170.4	168.7	164.1	156.4	156.2	157.2		188.8	284.3	330.7
of which: external 3/	...	86.0	106.8	97.3	94.7	95.0	92.1	94.0	95.4		108.8	160.4	192.8
Debt service-to-revenue and grants ratio (in percent) 4/	7.5	11.0	13.6	25.9	13.0	10.6	9.4	7.0	6.1		7.1	11.2	14.3
Debt service-to-revenue ratio (in percent) 4/	9.6	13.7	16.5	31.5	16.3	13.1	11.6	8.5	7.4		8.4	12.7	15.7
Primary deficit that stabilizes the debt-to-GDP ratio	3.2	1.1	-3.1	-1.9	2.4	2.5	2.6	2.5	2.3		2.3	2.9	3.3
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	6.6	4.0	4.0	5.2	5.9	6.1	6.2	6.0	6.0	5.6	6.0	6.0	6.0
Average nominal interest rate on forex debt (in percent)	1.0	1.0	1.2	1.6	1.7	2.0	2.2	2.4	2.5	1.8	3.0	3.3	3.4
Average real interest rate on domestic debt (in percent)	5.9	7.4	2.9	5.2	0.8	-0.5	-0.9	-1.4	-1.4	1.0	-1.0	-1.0	-1.6
Real exchange rate depreciation (in percent, + indicates depreciation)	-4.3	11.1	13.4	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-0.8	-1.7	1.6	2.6	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.2
Growth of real primary spending (deflated by GDP deflator, in percent)	15.0	-12.4	-3.5	15.9	12.2	10.5	10.0	5.4	7.3	8.4	5.1	3.8	5.2
Grant element of new external borrowing (in percent)	...	...	35.3	25.2	25.1	22.9	22.0	21.6	21.3	25.4	20.8	18.4	18.1

Sources: Country authorities; and staff estimates and projections.
1/ Central government gross debt
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A2.3. Burkina Faso: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35

(In percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to GDP ratio								
Baseline	17	17	17	18	18	18	**22**	34
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	17	18	20	23	24	26	**26**	38
A2. New public sector loans on less favorable terms in 2015-2035 2	17	17	18	19	20	21	**28**	50
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	17	17	17	18	18	19	**22**	35
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	17	20	24	25	25	25	**26**	36
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	17	18	19	20	20	21	**24**	39
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	17	18	19	20	20	21	**23**	35
B5. Combination of B1-B4 using one-half standard deviation shocks	17	17	19	19	20	20	**23**	36
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	17	24	24	25	25	26	**30**	49
PV of debt-to-exports ratio								
Baseline	68	66	67	71	73	75	**101**	193
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	68	71	81	91	99	105	**120**	213
A2. New public sector loans on less favorable terms in 2015-2035 2	68	68	71	78	83	88	**130**	280
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	68	66	67	71	73	75	**101**	193
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	68	88	123	127	129	131	**158**	257
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	68	66	67	71	73	75	**101**	193
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	68	69	76	80	82	84	**108**	195
B5. Combination of B1-B4 using one-half standard deviation shocks	68	67	69	73	75	78	**102**	193
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	68	66	67	71	73	75	**101**	193
PV of debt-to-revenue ratio								
Baseline	107	97	95	95	92	94	**109**	193
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	107	105	114	122	125	131	**130**	212
A2. New public sector loans on less favorable terms in 2015-2035 2	107	100	100	105	105	109	**140**	279
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	107	99	97	98	95	96	**112**	198
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	107	113	136	133	127	127	**132**	199
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	107	103	106	107	104	105	**122**	216
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	107	102	108	108	104	105	**116**	195
B5. Combination of B1-B4 using one-half standard deviation shocks	107	100	105	105	101	103	**117**	204
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	107	138	134	134	131	133	**154**	272

Table A2.3. Burkina Faso: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	4	4	5	5	5	5	**7**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	4	4	5	5	6	6	**8**	15
A2. New public sector loans on less favorable terms in 2015-2035 2	4	4	4	5	5	5	**8**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	4	4	5	5	5	5	**7**	15
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	4	5	6	8	8	8	**12**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	4	4	5	5	5	5	**7**	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	4	4	5	5	6	5	**8**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	5	5	5	5	**7**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	4	4	5	5	5	5	**7**	15
Debt service-to-revenue ratio								
Baseline	7	6	6	7	7	6	**8**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	7	6	7	7	7	7	**9**	15
A2. New public sector loans on less favorable terms in 2015-2035 2	7	6	6	7	7	7	**9**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	7	6	7	7	7	6	**8**	15
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	7	6	7	8	8	7	**10**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	7	7	7	8	7	7	**9**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	7	6	7	7	7	7	**8**	15
B5. Combination of B1-B4 using one-half standard deviation shocks	7	6	7	7	7	7	**8**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	7	9	9	10	9	9	**11**	21
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	17	17	17	17	17	17	**17**	17

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A2.4. Burkina Faso: Sensitivity Analysis for Key Indicators of Public Debt, 2015–35

	Projections											
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2035
PV of Debt-to-GDP Ratio												
Baseline	26	29	30	30	30	31	31	32	34	35	37	59
A. Alternative scenarios												
A1. Real GDP growth and primary balance are at historical averages	26	29	29	28	27	27	26	26	26	26	25	23
A2. Primary balance is unchanged from 2015	26	30	29	28	28	27	27	26	26	26	26	24
A3. Permanently lower GDP growth 1/	26	30	31	31	32	32	33	35	37	40	42	74
B. Bound tests												
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	26	31	32	32	33	33	34	36	38	40	42	67
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	26	35	40	40	40	39	40	40	42	43	45	64
B3. Combination of B1-B2 using one half standard deviation shocks	26	32	35	35	35	35	36	37	38	40	42	64
B4. One-time 30 percent real depreciation in 2016	26	37	36	36	36	35	36	36	38	39	41	62
B5. 10 percent of GDP increase in other debt-creating flows in 2016	26	39	39	38	38	38	38	39	40	42	44	63
PV of Debt-to-Revenue Ratio 2/												
Baseline	133	140	135	132	126	129	130	134	140	148	160	300
A. Alternative scenarios												
A1. Real GDP growth and primary balance are at historical averages	133	139	130	123	114	112	109	108	107	107	109	117
A2. Primary balance is unchanged from 2015	133	140	131	124	115	113	111	109	109	109	111	120
A3. Permanently lower GDP growth 1/	133	144	138	137	132	135	138	145	153	164	180	374
B. Bound tests												
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	133	145	142	141	137	140	142	148	156	166	180	338
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	133	167	181	175	165	166	165	168	173	180	191	323
B3. Combination of B1-B2 using one half standard deviation shocks	133	154	157	153	146	148	149	153	159	167	180	323
B4. One-time 30 percent real depreciation in 2016	133	173	163	157	149	149	149	152	157	164	176	316
B5. 10 percent of GDP increase in other debt-creating flows in 2016	133	183	173	167	159	159	159	162	167	174	186	319
Debt Service-to-Revenue Ratio 2/												
Baseline	14	26	13	11	9	7	6	6	6	7	7	14
A. Alternative scenarios												
A1. Real GDP growth and primary balance are at historical averages	14	26	13	10	9	6	5	5	5	4	4	2
A2. Primary balance is unchanged from 2015	14	26	13	10	9	6	5	5	5	5	4	2
A3. Permanently lower GDP growth 1/	14	26	13	11	10	7	7	7	7	7	8	19
B. Bound tests												
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	14	26	13	11	10	8	7	7	7	7	8	17
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	14	26	14	15	13	9	8	7	7	9	10	17
B3. Combination of B1-B2 using one half standard deviation shocks	14	26	13	12	11	8	7	7	7	8	9	16
B4. One-time 30 percent real depreciation in 2016	14	27	15	13	12	10	9	9	9	10	11	23
B5. 10 percent of GDP increase in other debt-creating flows in 2016	14	26	15	16	11	9	7	7	7	9	9	16

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.